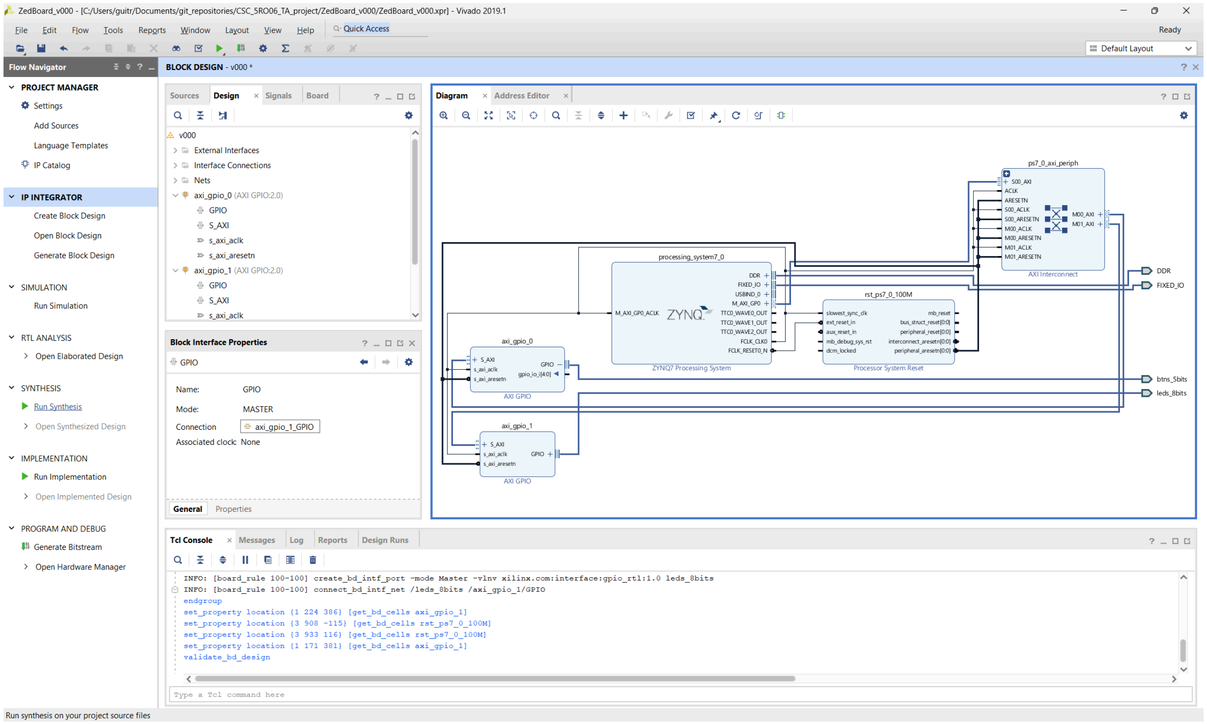
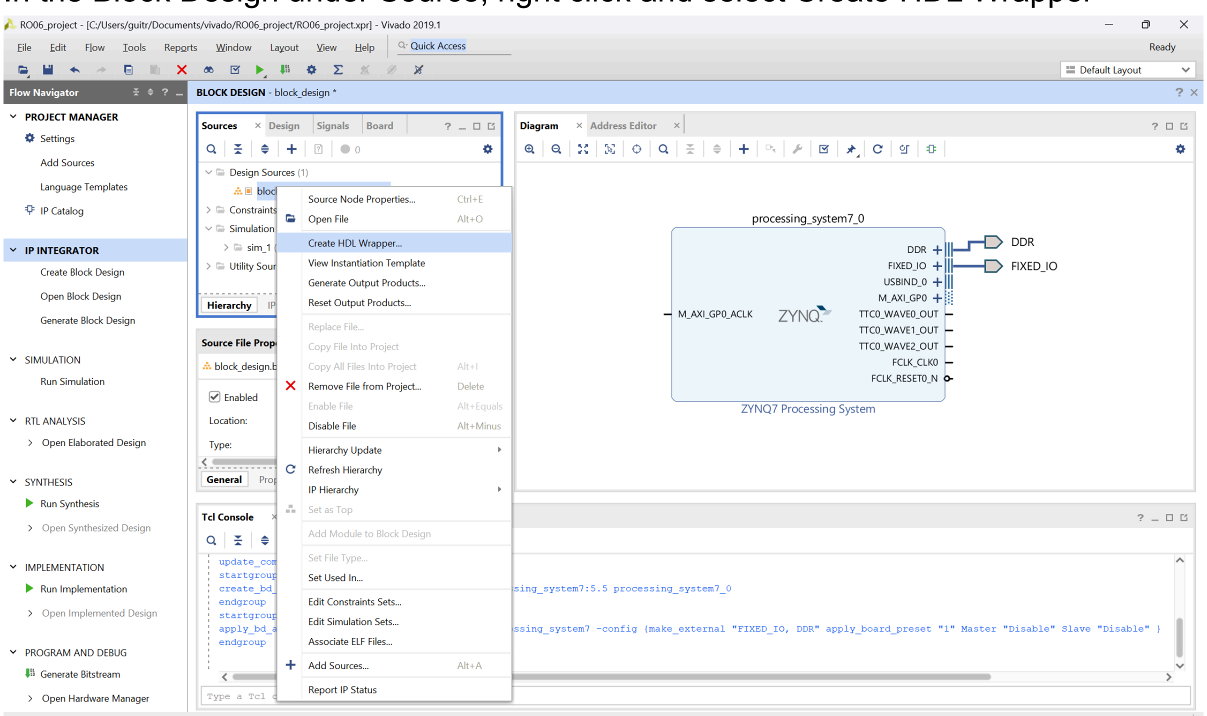
Vivado 2019.1 - Generate Bitstream

Create HDL Wrapper

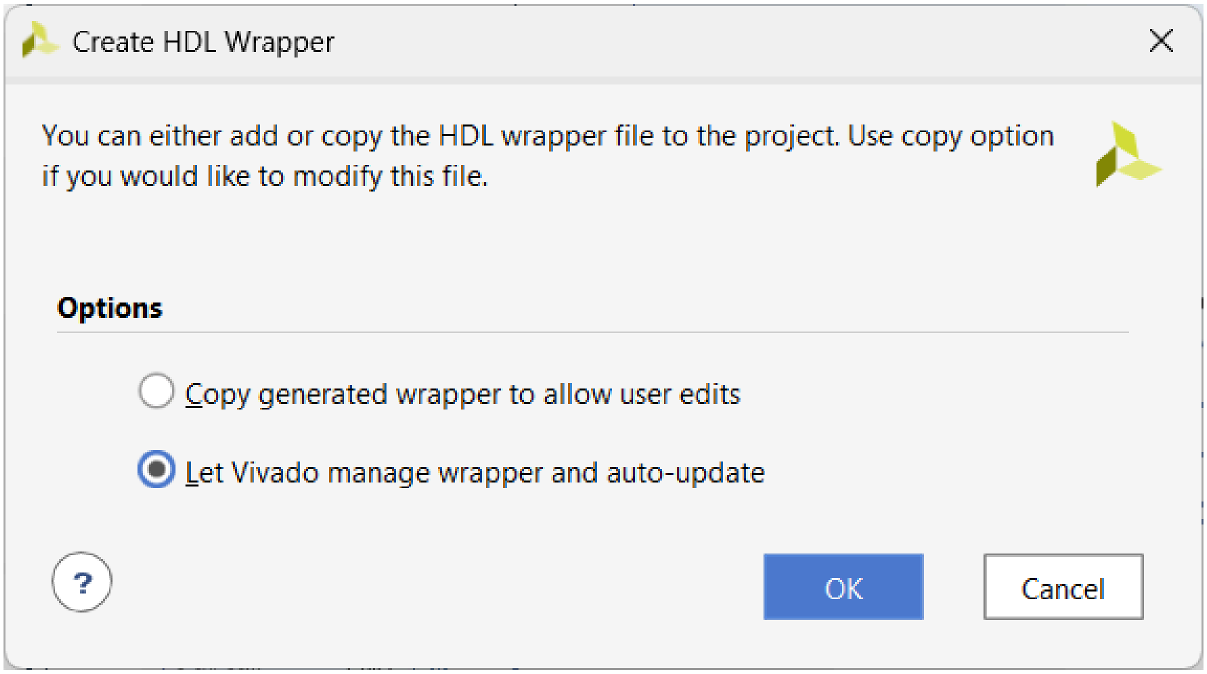
After generating the block design it is possible to Run Syntheses under Synthesis menu.



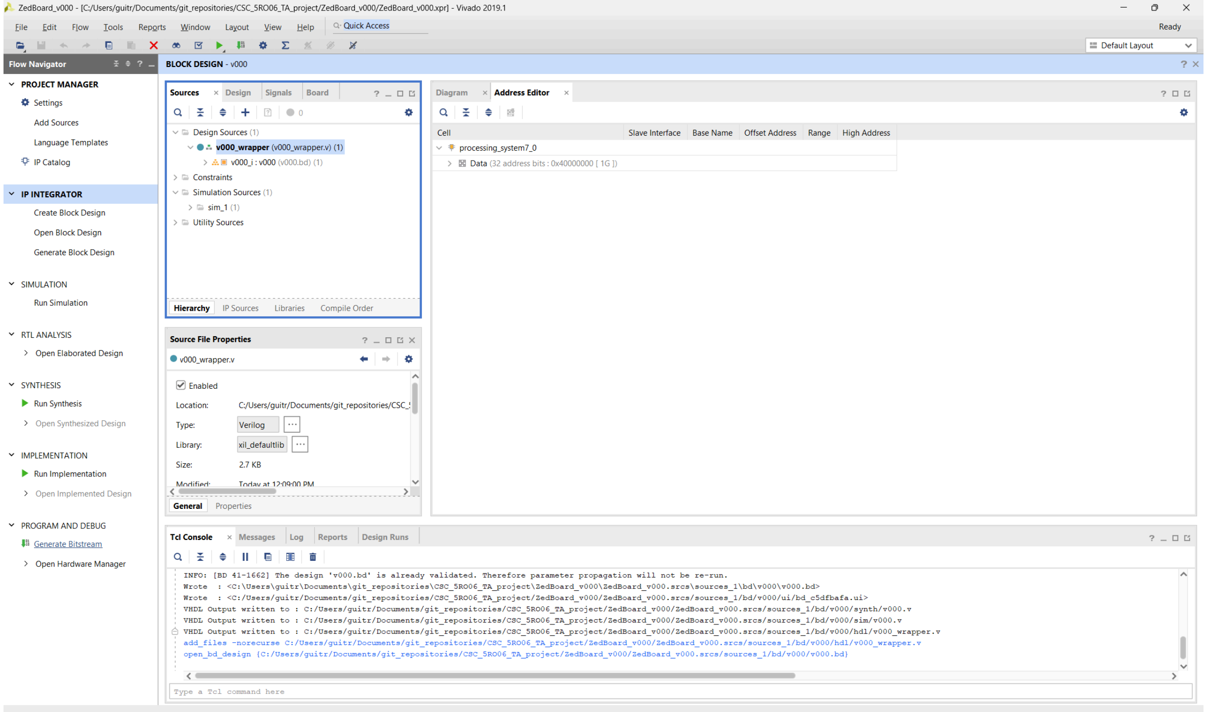
In the Block Design under Source, right click and select Create HDL Wrapper

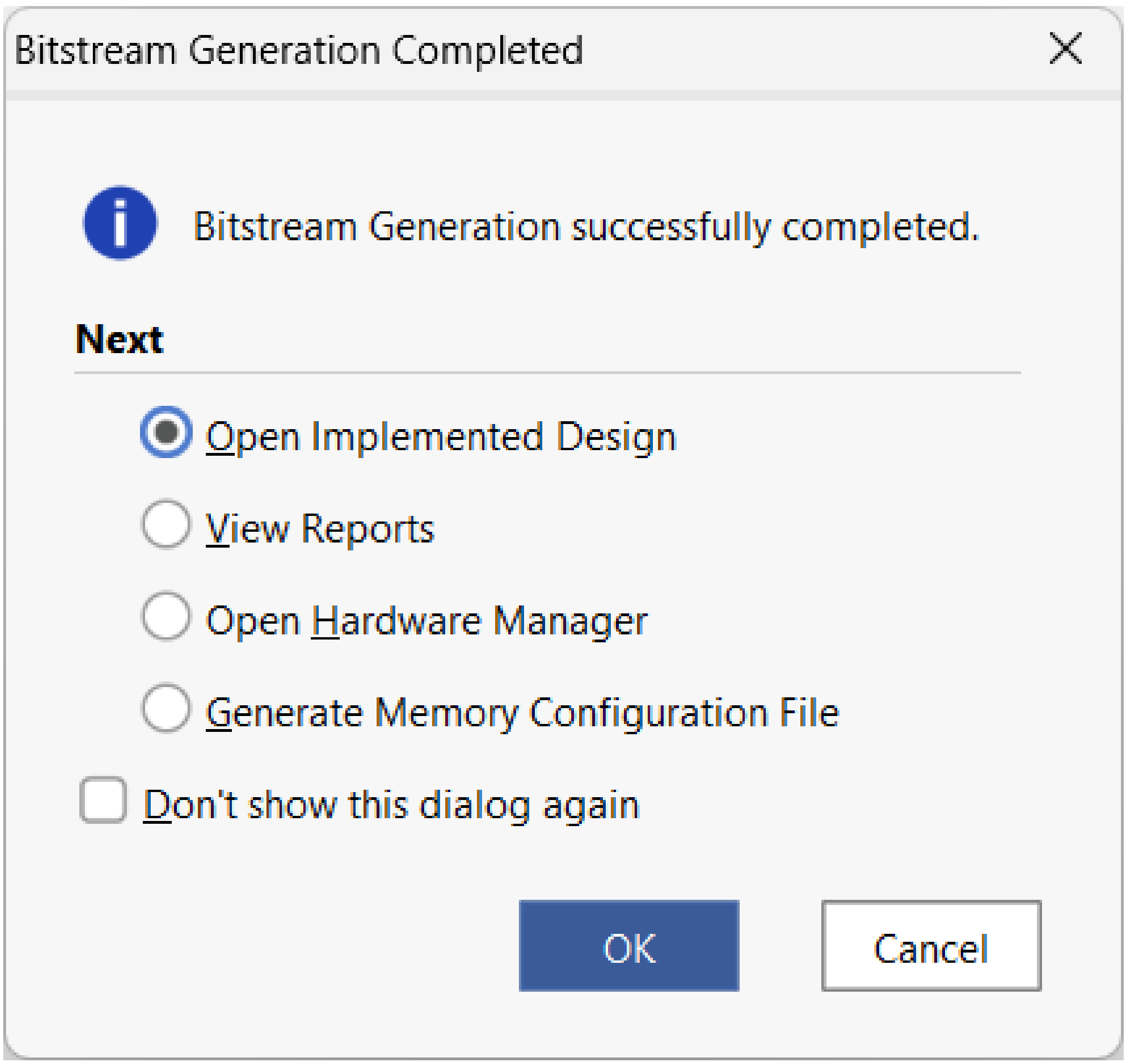


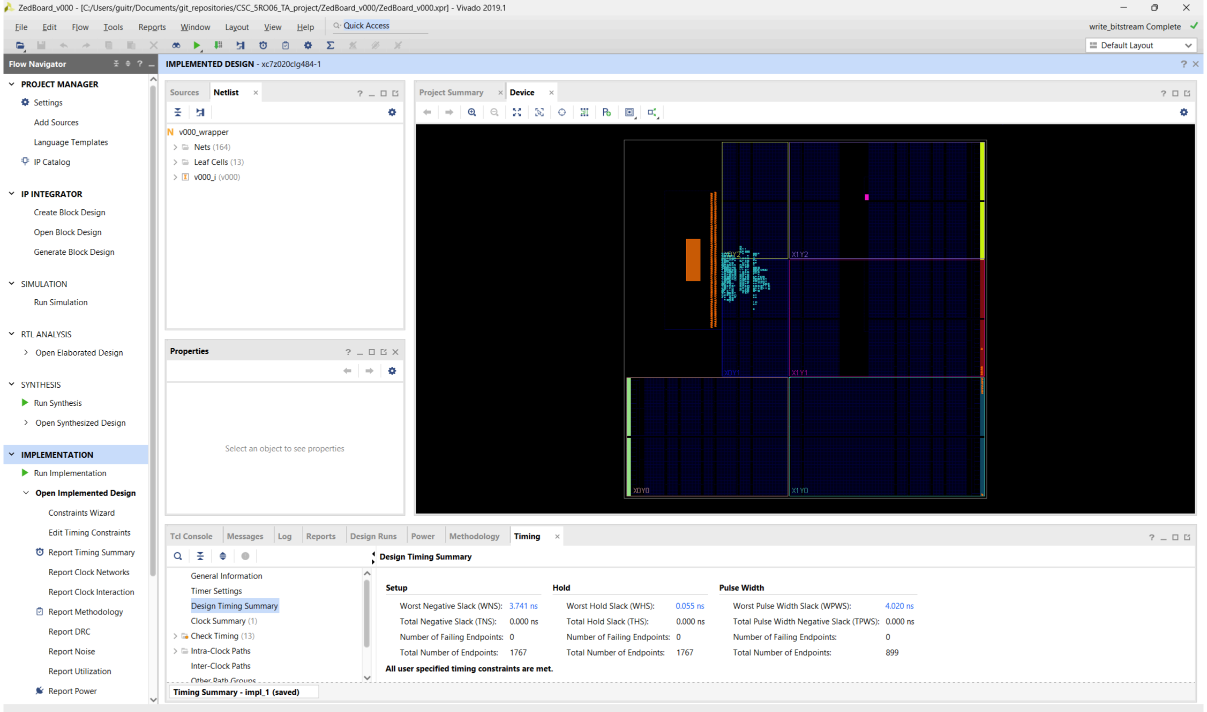
Let Vivado decide what to do.



Generate Bitstream


ZedBoard_v000 - [C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.xpr] - Vivado 2019.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
Ready
Default Layout
Flow Navigator
BLOCK DESIGN - v000
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
Create Block Design
Open Block Design
Generate Block Design
SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
SYNTHESIS
Run Synthesis
Open Synthesized Design
IMPLEMENTATION
Run Implementation
Open Implemented Design
PROGRAM AND DEBUG
Generate Bitstream
Open Hardware Manager
Sources
Design
Signals
Board
Design Sources (1)
v000_wrapper (v000_wrapper.v) (1)
v000_i : v000 (v000.bd) (1)
Constraints
Simulation Sources (1)
sim_1 (1)
Utility Sources
Hierarchy
IP Sources
Libraries
Compile Order
Source File Properties
v000_wrapper.v
Enabled
Location: C:/Users/guitr/Documents/git_repositories/CSC_!
Type: Verilog
Library: xil_defaultlib
Size: 2.7 KB
Modified: Today at 12:09:00 PM
General
Properties
Diagram
Address Editor
Cell
Slave Interface
Base Name
Offset Address
Range
High Address
processing_system7_0
Data (32 address bits : 0x40000000 [1G])
Tcl Console
Messages
Log
Reports
Design Runs
INFO: [BD 41-1662] The design 'v000.bd' is already validated. Therefore parameter propagation will not be re-run.
Wrote : <C:\Users\guitr\Documents\git_repositories\CSC_5RO06_TA_project\ZedBoard_v000\ZedBoard_v000.srcs\sources_1\bd\v000\v000.bd>
Wrote : <C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/ui/bd_c5dfbafa.ui>
VHDL Output written to : C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/synth/v000.v
VHDL Output written to : C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/sim/v000.v
VHDL Output written to : C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/hdl/v000_wrapper.v
add_files -norecurse C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/hdl/v000_wrapper.v
open_bd_design {C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.srcs/sources_1/bd/v000/v000.bd}
Type a Tcl command here


Bitstream Generation Completed
Bitstream Generation successfully completed.
Next
Open Implemented Design
View Reports
Open Hardware Manager
Generate Memory Configuration File
Don't show this dialog again
OK
Cancel


ZedBoard_v000 - [C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000/ZedBoard_v000.xpr] - Vivado 2019.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
write_bitstream Complete
Default Layout
Flow Navigator
IMPLEMENTED DESIGN - xc7z020clg484-1
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
Create Block Design
Open Block Design
Generate Block Design
SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
SYNTHESIS
Run Synthesis
Open Synthesized Design
IMPLEMENTATION
Run Implementation
Open Implemented Design
Constraints Wizard
Edit Timing Constraints
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Noise
Report Utilization
Report Power
Sources
Netlist
v000_wrapper
Nets (164)
Leaf Cells (13)
v000_i (v000)
Properties
Select an object to see properties
Project Summary
Device
X1Y2
X1Y1
X0Y0
X1Y0
Tcl Console
Messages
Log
Reports
Design Runs
Power
Methodology
Timing
General Information
Timer Settings
Design Timing Summary
Clock Summary (1)
Check Timing (13)
Intra-Clock Paths
Inter-Clock Paths
Other Path Groups
Design Timing Summary
Setup
Worst Negative Slack (WNS): 3.741 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 1767
Hold
Worst Hold Slack (WHS): 0.055 ns
Total Hold Slack (THS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 1767
Pulse Width
Worst Pulse Width Slack (WPWS): 4.020 ns
Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 899
All user specified timing constraints are met.
Timing Summary - impl_1 (saved)